

Visual Semantic Processing in Primate Frontotemporal Cognitive Networks and Artificial Systems

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Abstract

Visual perception involves both the extraction of physical features and the abstraction of semantic meaning, yet the neural mechanisms by which the primate brain derives semantic information remain incompletely understood. We recorded neuronal activity simultaneously in the inferior temporal cortex (ITC) and ventrolateral prefrontal cortex (vIPFC) of macaques viewing natural and artificial objects. Mid-level semantic distinctions (e.g., faces versus bodies) were primarily represented in ITC populations, whereas high-level abstractions (e.g., animate versus inanimate) mainly resolved in PFC. Functional connectivity analyses revealed bidirectional information flow between these regions, indicating dynamic exchange of object features. Notably, encoding axes for mid- and high-level semantics were orthogonally aligned in PFC early after stimulus onset, suggesting distinct neural subspaces support independent processing of different attributes of the visual input. Finally, state-of-the-art deep learning models exhibited progressive abstraction along feed-forward layers, paralleling hierarchical processing observed in the primate brain. Together, these findings clarify the division of information processing between ITC and PFC, the temporal sequence of semantic processing, and the mechanisms underlying the perceptual independence of different object attributes.

Introduction

Primates exhibit remarkable proficiency in recognizing and categorizing objects rapidly and accurately, a capacity critical for survival, navigation in complex environments, and social interaction¹⁻³. Psychophysical studies indicate that object recognition unfolds across multiple stages rather than occurring instantaneously^{4,5}. Early stages involve extraction of basic visual features, such as shape, texture, and coarse configurations, which support rapid discrimination between objects^{3,4,6}. More abstract, conceptual categorizations typically arise later and recruit additional cognitive resources, reflecting the hierarchical and interactive nature of visual processing^{3,4}.

Behavioral evidence shows that mid-level semantic distinctions, such as faces versus bodies, can be extracted quickly, whereas higher-order categories, including animacy, often require more processing time and involve broader neural networks^{4,5}. Whether this temporal sequence reflects a strict dependency of higher-level abstractions on earlier perceptual computations or an evolutionary/experiential prioritization of certain features remains unresolved⁴.

At the neural level, this hierarchy involves interactions between the inferior temporal cortex (ITC) and prefrontal cortex (PFC), regions critically implicated in object representation and decision-making^{2,7-11}. ITC has been traditionally associated with encoding visual and physical object features¹²⁻¹⁵, whereas PFC is linked to abstract, context-dependent categorization¹⁰⁻¹⁶. Debate persists over whether inferior temporal cortex (ITC) encodes semantic information directly or whether semantics emerge only after ITC outputs are integrated by higher-order regions that incorporate prior knowledge (e.g., PFC)^{9,10,12-20}. One view holds that mid-level semantic information is resolved early in ITC, consistent with a basic-level advantage, and then conveyed feed-forward to prefrontal regions⁷. An alternative account proposes that ITC primarily represents visual, non-semantic features, with abstraction arising downstream through integrative processes that leverage prior knowledge and task demands^{9,10,12-20}. Although current models posit bidirectional information flow between ITC and PFC^{21,22}, the precise timing, directionality, and representational content of these exchanges remain unresolved. Consequently, Direct evidence demonstrating how abstract representations arise within this circuit is limited.

Here, we address these questions by recording neuronal activity simultaneously in macaque ITC and PFC during passive viewing of diverse visual objects. We find that mid-level semantic information emerges earlier than high-level categories, consistent with hierarchical processing. ITC recurrent circuitry appears to support mid-level distinctions, while PFC contributes to higher-level categorizations. Enhanced functional connectivity between the regions following stimulus onset suggests dynamic, bidirectional information transfer. Furthermore, orthogonal alignment of mid- and high-level encoding axes in PFC neural space may provide a mechanism for maintaining separable representations of distinct object features. Finally, we show that progressive abstraction in state-of-the-art deep neural networks mirrors the hierarchical progression observed in primates, though the underlying computational mechanisms differ substantially. Collectively, these results provide new insights into the circuit-level basis of semantic processing in the primate brain.

Results

Neurophysiology and Behavior

We simultaneously recorded the neuronal activity of the inferior temporal cortex (ITC) and the prefrontal cortex (PFC) while macaque monkeys viewed a series of visual stimuli comprising isolated grayscale natural and artificial objects of several categories (Fig. 1a). Given that the spatial frequency profile, a basic visual feature, can influence category processing in ITC^{7,23}, we matched it across the different stimulus categories (see Methods). Following this adjustment, no significant difference was detected among categories for any spatial frequency (all comparisons showed a $p > 0.05$; Supplementary Fig. 1). The stimulus set was designed to explore both mid-level and high-level semantic distinctions, spanning conceptual categories such as animate versus inanimate objects, and faces versus bodies. To cover a broad range of categories for each high-level abstraction class, we employed a rapid serial visual presentation (RSVP) paradigm to maximize trial collection within a feasible timeframe. In each trial, the stimulus appeared centrally (7° visual field) for 80 ms, followed by a 400 ms blank screen (Fig. 1a). The animal received a juice reward for continuous central fixation during random periods ranging from 2–4 seconds. Fig. 1b illustrates the hypothetical semantic hierarchy, which is consistent with a previous study⁷. For high-level abstraction, stimuli were broadly divided into animate and inanimate objects. At the mid-level, animate objects were subdivided into primate or non-primate, and face or body, while inanimate objects were natural or artificial. Recognizing that some classifications may rely on human-level knowledge, we focused the remainder of this study on the animate vs. inanimate distinction for high-level abstraction and the face vs. body distinction for mid-level abstraction. This clearly defined stimulus set allowed us to explore the temporal dynamics of object representation in the ITC-PFC circuit.

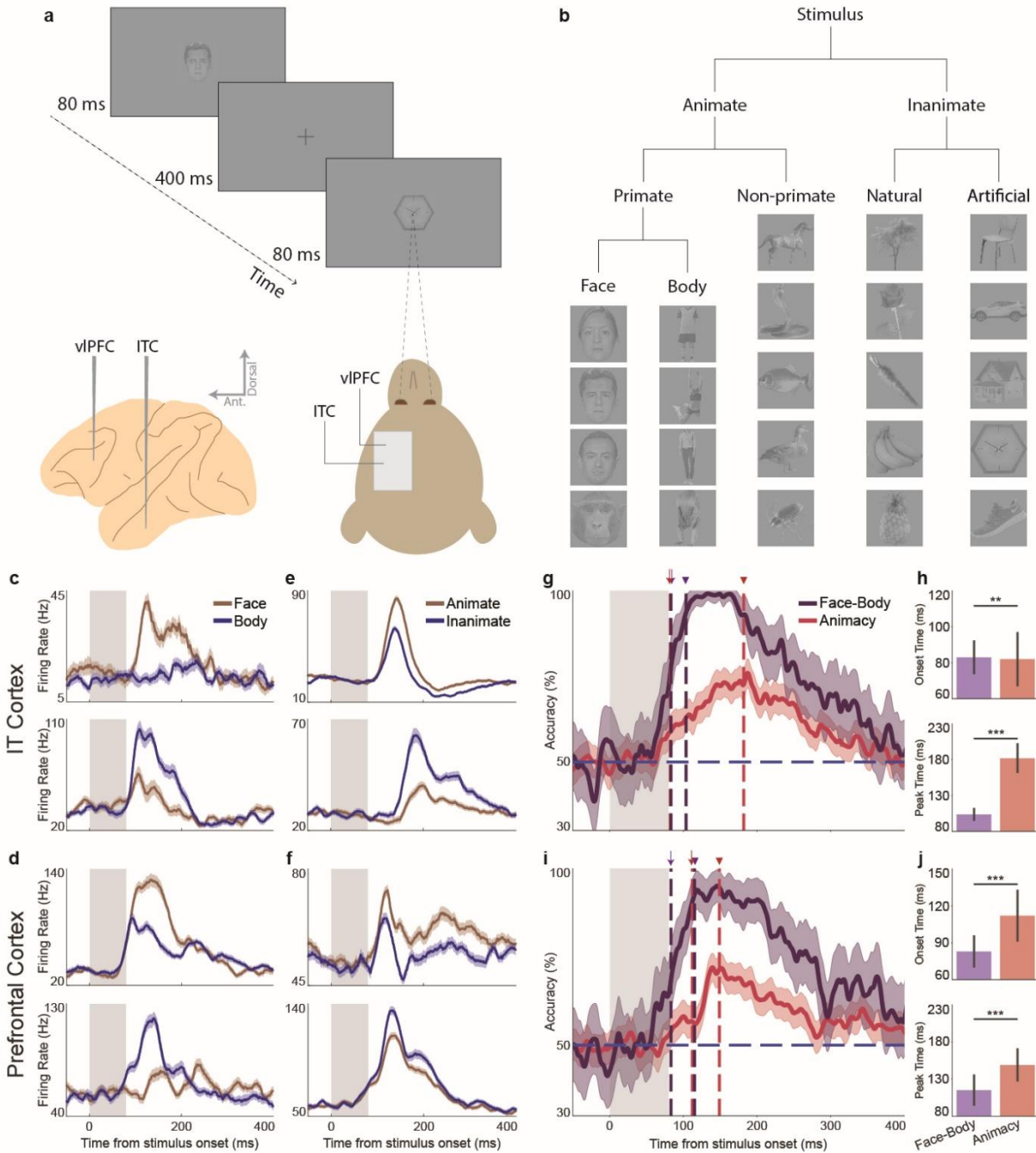


Figure 1. Experimental design, theoretical framework, and regional information representation.

(a) Schematic of the experimental paradigm. The animal was trained to fixate centrally and view visual objects to receive juice rewards. In each trial, a stimulus was presented within the central 7° of the visual field for 80 ms, followed by a 400 ms blank interval. Neuronal activity was simultaneously recorded from ITC and vIPFC for offline analysis. Bottom left, schematic of recording sites. (b) Semantic hierarchy of the visual stimuli, illustrating categorical distinctions: animate versus inanimate (high-level) and face versus body (mid-level). (c-f) Peri-stimulus time histograms (PSTHs) of example units selective for face (c,d, upper), body (c,d, lower), animate (e,f, upper), and inanimate (e,f, lower) stimuli in ITC (c,e) and PFC (d,f). Solid lines indicate mean firing rates; shaded areas denote SEM. The gray bar (0–80 ms) marks stimulus presentation. (g,i) Time course of decoding accuracy for classifiers

trained and tested on neural population activity from ITC (**g**) and PFC (**i**) to discriminate face versus body and animate versus inanimate categories. Solid lines represent mean accuracy; shaded areas denote SD. The gray bar (0–80 ms) indicates stimulus presentation. Dashed lines (arrows and arrowheads) indicate median onset and peak latencies, respectively. (**h,j**) Statistical comparison of onset (top) and peak (bottom) latencies for classifiers in **g** and **i**, shown for ITC (**h**) and PFC (**j**). Bars indicate medians; error bars denote SD. Significance was assessed using permutation tests (** $P < 0.01$, *** $P < 0.001$). ITC: inferior temporal cortex; vIPFC: ventrolateral prefrontal cortex; PSTH: peri-stimulus time histogram.

Basic Level Advantage of ITC and PFC

Fig. 1c–f presents sample units exhibiting systematic selectivity for faces (upper panels) and bodies (lower panels) (Fig. 1c,d) or animates (upper panels) and inanimates (lower panels) (Fig. 1e,f) in ITC (Fig. 1c,e) and PFC (Fig. 1d,f), respectively. To determine the temporal dynamics of semantic representation in each region, we trained linear discriminant analysis (LDA) classifiers to distinguish face/body and animate/inanimate objects using population neuronal activity (Fig. 1g,i).

We found that mid-level semantic information emerged earlier than high-level information in ITC (median $\pm$ SD, ms; onset: mid-level = 83.0 ± 8.8 , high-level = 82.0 ± 14.4 , $P = 0.002$; peak: mid-level = 103.5 ± 7.6 , high-level = 182.0 ± 19.4 , $P < 1 \times 10^{-4}$; permutation test; Fig. 1 g,h). Notably, a similar temporal hierarchy was observed in PFC population activity (median $\pm$ SD, ms; onset: mid-level = 83.0 ± 12.3 , high-level = 112.0 ± 20.1 , $P < 1 \times 10^{-4}$; peak: mid-level = 115.0 ± 19.7 , high-level = 149.0 ± 20.8 , $P < 1 \times 10^{-4}$; permutation test; Fig. 1i,j). These findings are consistent with previous observations on the early representation of mid-level information in ITC⁷. Furthermore, considering the PFC's engagement in psychophysical tasks requiring perceptual responses¹⁶, these observations suggest a rationale for faster categorization of basic-level information at the behavioral or perceptual level, aligning with prior reports^{4,5}.

ITC and PFC Processing of Semantic Attributes

We next explored how the ITC-PFC circuit processes these two semantic levels. Following previous work, we reasoned that the speed of accumulation of perceptual evidence over time, rather than onset or peak times alone, suggests the time-course of information representation^{4,24–26} (see exemplar-based random walk model, Fig. 2a). To capture the time required for full problem representation, we used solution time, defined as the difference between the onset and peak times (Fig. 2b), which can suggest the ability to solve the information.

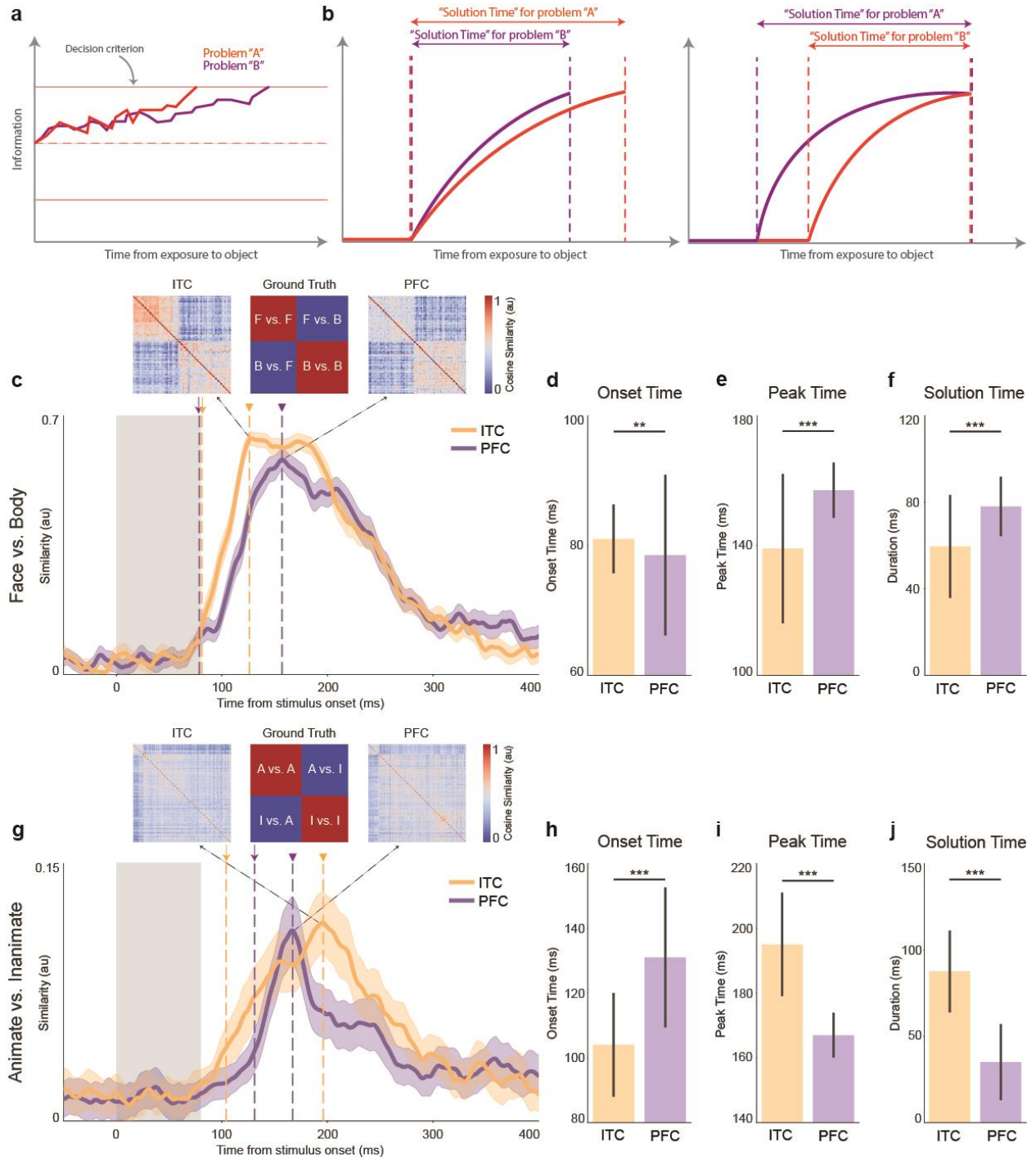


Figure 2. Temporal dynamics of semantic information in the ITC–PFC circuit.

(a,b) Schematic illustrations of the theoretical framework, including the random-walk model (a) and definition of solution time (b). (c,g, bottom) Time course of similarity to ground truth for population activity in ITC and PFC for face versus body (c) and animate versus inanimate (g) distinctions. Solid lines indicate mean similarity; shaded areas denote SD. The gray bar (0–80 ms) marks stimulus presentation. Dashed lines (arrows and arrowheads) indicate median onset and peak latencies, respectively. (c,g, top) Representational similarity matrices (RSMs) at peak similarity for ITC (left) and PFC (right), shown alongside ground truth (center) for face–body (c) and animate–

inanimate (**g**) conditions. Warmer colors indicate higher cosine similarity. (**d-f, h-j**) Statistical comparisons of onset (**d,h**), peak (**e,i**), and solution (**f,j**) times derived from **c** and **g**, for face–body (**d-f**) and animate–inanimate (**h-j**) distinctions. Bars represent medians; error bars denote SD. Significance was assessed using permutation tests (***) $p < 0.001$. ITC: inferior temporal cortex; PFC: prefrontal cortex; RSM: representational similarity matrix.

We tracked the evolution of mid-level (face vs. body) and high-level (animate vs. inanimate) information using Representational Similarity Analysis (RSA)²⁷. Ground truth Representational Similarity Matrices (RSMs) were created for each distinction, and instantaneous RSMs from ITC and PFC population activity were compared to the ground truth (Fig. 2c,g).

For mid-level (face vs. body) information (Fig. 2c-d): despite a negligible onset difference (median $\pm$ SD in ms; ITC = 81.00 ± 5.64 , PFC = 80.00 ± 11.65 , $p = 0.004$; permutation test; Fig. 2c,d), mid-level information peaked earlier (ITC = 135.00 ± 21.73 , PFC = 157.00 ± 7.51 , $p < 1e-4$; permutation test; Fig. 2c,e) and was solved faster (ITC = 57.00 ± 22.12 , PFC = 78.00 ± 12.44 , $p < 1e-4$; permutation test; Fig. 2c,f) in the ITC compared to PFC. This observation is consistent with a major theory suggesting a role for ITC in the semantic perception of faces in primates^{28–30}.

For high-level (animate vs. inanimate) information (Fig. 2g–j): This pattern was reversed. High-level representations initiated later in the PFC (ITC = 104.00 ± 15.62 , PFC = 131.00 ± 21.17 , $p < 1e-4$; permutation test; Fig. 2g,h), but reached full representation earlier than the ITC (ITC = 195.00 ± 15.59 , PFC = 167.00 ± 6.65 , $p < 1e-4$; permutation test; Fig. 2g,i). These results suggest greater capability of the PFC population in solving the high-level abstractions, evidenced by shorter solution times in PFC compared to ITC (ITC = 87.50 ± 23.02 , PFC = 35.00 ± 21.35 , $p < 1e-4$; permutation test; Fig. 2g,j). This finding is consistent with the extensive literature suggesting roles for PFC in high-level categorizations and perceptions^{9,10,16,19,31}. This notion also suggests that mid-level information appears earlier in PFC (earlier onset and peak times; Fig. 1i,j) because it is received via the feed-forward pathway, whereas high-level information, which is not readily available in the initial feed-forward input, is solved with more timing within the PFC circuit (Fig. 2g–j).

Confirmation with Encoding Models

Objects in the two categories at both semantic levels differed in their physical features. We quantified several low-level visual properties (including circularity, elongation, spikiness, contrast, luminance, object area, and the first two principal components of the grayscale images; see Methods), as either region could respond to these features^{12–15,20,32} independent of semantic processing. Supplementary Fig. 2a shows that the 20 most circular objects contain a greater number of inanimate and some faces, while the 20 most spiky objects are predominantly animate (include several bodies, no faces; Supplementary Fig. 2b). This striking difference prompted a need to factor out the contribution of physical features.

We used Generalized Linear Models (GLMs) to predict instantaneous neural responses from a set of semantic and physical features. We compute object residual encoding models by first fitting a GLM that deliberately excludes a target feature, like animacy, which forces the model's predictions to be blind to that signal. The residuals are then calculated for each stimulus as the simple difference between the actual neural response and the model's prediction, thereby computationally isolating the enhanced signal of the omitted feature in the resulting residual values. We then computed the object residuals for each semantic level, which suggests the near-exclusive contribution of the specific semantic feature, similar to a previous study³³ (see Methods for details). Fig. 3a–d displays the object residuals for sample units, illustrating category-selective activity after accounting for low-level features (sample unit in ITC: face selective: Fig. 1a, upper panels, body selective: Fig. 1a, lower panels, animate selective: Fig. 1c, upper panels, inanimate selective: Fig. 1c, lower panels, sample unit in PFC: face selective: Fig. 1b, upper panels, body selective: Fig. 1b, lower panels, animate selective: Fig. 1d, upper panels, inanimate selective: Fig. 1d, lower panels).

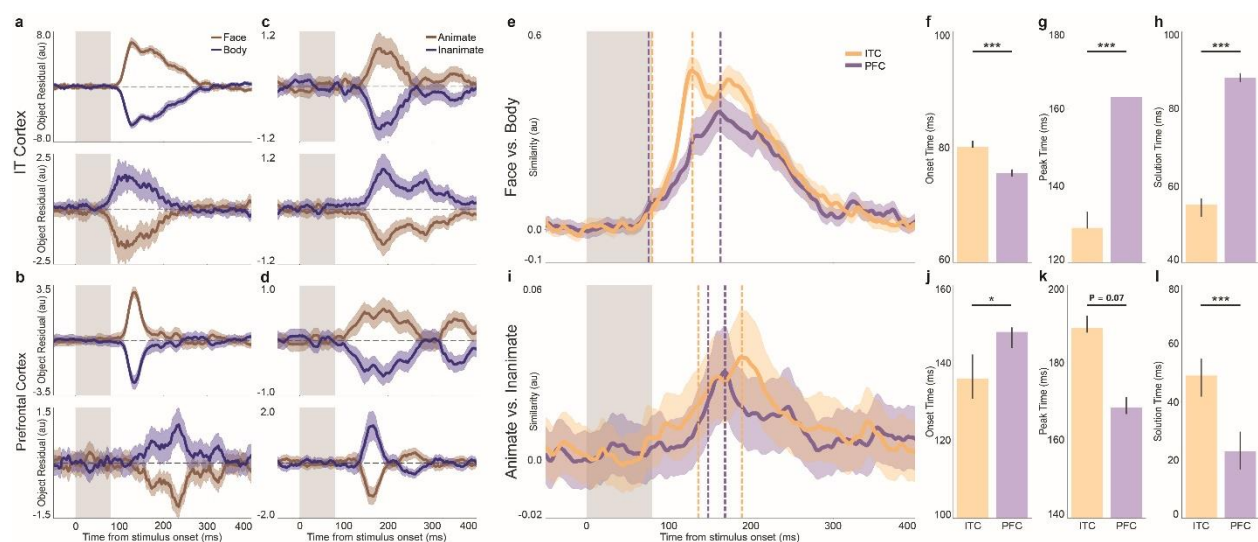


Figure 3. Temporal dynamics of enhanced semantic information in the ITC–PFC circuit.

(a–d) Object residuals for example units selective for face (a,b, upper), body (a,b, lower), animate (c,d, upper), and inanimate (c,d, lower) stimuli in ITC (a,c) and PFC (b,d). Solid lines indicate mean residuals; shaded areas denote SEM. The gray bar (0–80 ms) marks stimulus presentation. (e,i) Time course of similarity to ground truth for population residuals in ITC and PFC for face versus body (e) and animate versus inanimate (i) distinctions. Solid lines represent mean similarity; shaded areas denote SD. The gray bar (0–80 ms) indicates stimulus presentation. Dashed lines (arrows and arrowheads) indicate median onset and peak latencies, respectively. (f–h,j–l) Statistical comparisons of onset (f,j), peak (g,k), and solution (h,l) times derived from e and i, for face–body (f–h) and animate–inanimate (j–l) distinctions. Bars represent medians; error bars denote 95% confidence intervals. Significance was assessed using permutation tests (* $P < 0.05$, *** $P < 0.001$). CI: confidence interval; ITC: inferior temporal cortex; PFC: prefrontal cortex.

These object residuals were then subjected to the RSA procedure (as in Fig. 2c–j), and the results were largely consistent with the earlier observations. Specifically: For face-body separation (mid-level): The separation initiated earlier in PFC (median $\pm$ SD in ms; ITC = 80.00 ± 8.98 , PFC = 75.50 ± 10.66 , $p < 1e-4$; permutation test; Fig. 3e,f), but peaked earlier (ITC = 129.00 ± 21.98 , PFC = 163.00 ± 12.99 , $p < 1e-4$; permutation test; Fig. 3e,g) and was solved faster (ITC = 55.00 ± 23.46 , PFC = 88.00 ± 16.01 , $p < 1e-4$; permutation test; Fig. 3e,h) in the ITC population. For animacy information (high-level): This information had an earlier onset in ITC (ITC = 136.00 ± 40.00 , PFC = 148.00 ± 39.92 , $p = 0.04$; permutation test; Fig. 3i,j), but an earlier peak representation (ITC = 189.00 ± 32.37 , PFC = 168.50 ± 39.28 , $p = 0.07$; permutation test; Fig. 3i,k) and shorter solution time (ITC = 49.00 ± 40.81 , PFC = 23.00 ± 44.92 , $p = 0.0007$; permutation test; Fig. 3i,l) in the PFC population.

By analytically controlling for several low-level features (in addition to matching spatial frequency and color in the stimulus design), we provide near-confirmatory evidence that the exclusive evaluation of semantic features suggests roles for ITC in processing mid-level, rapidly-categorized semantic information and PFC for high-level, perceptually-advanced abstractions in the primate brain. It is important to note that other visual features remain unaccounted for, suggesting a cautious interpretation of these findings.

Information Transfer Between ITC and PFC

Given that both regions are implicated in object processing and dynamically transfer information^{21,34}, we examined the functional connectivity using Canonical Correlation Analysis (CCA)³⁵ (Fig. 4a). CCA identifies dimensions of the two neural spaces (ITC and PFC) where activity is maximally correlated. We observed a pronounced and prolonged enhancement of functional connectivity between ITC and PFC following stimulus onset (Fig. 4b; thick and thin lines denote $p < 0.001$ and $p < 0.05$, respectively, measured by one-tailed permutation testing compared to baseline). This suggests a route for inter-areal information transfer.

If two neuronal populations are functionally connected, their representations are expected to be shared. Previous studies have shown feed-forward information flow from sensory regions such as ITC to associative areas including PFC^{11,34,36–38}, as well as feedback from prefrontal regions to sensory cortices under various conditions, including object recognition^{21,34,39,40}. Motivated by this bidirectional architecture, we applied a previously described approach, analogous to Granger causality, to quantify the extent to which activity in one region could be predicted from the past activity of the other^{41–43}. Specifically, we tested whether a region's representational similarity matrix (RSM) at a given time point could be predicted from earlier RSMs of the same region alone or from both regions over time. At each time point t , we fitted regression models to predict RSM_t using past RSMs from either the same area or from both areas; a reduction in prediction error when including RSMs from both regions indicates that the current population activity of one region is predictable from the past activity of the other (see Methods; Fig. 4c). Fig. 4d illustrates the time-resolved dynamics of information transfer. We detected bidirectional information flow, with the feed-forward (ITC-to-PFC) transfer peaking

earlier than the feedback (PFC-to-ITC) transfer (median $\pm$ SD in ms; ITC-to-PFC = 126.00 $\pm$ 10.32, PFC-to-ITC = 146.00 $\pm$ 21.07, $p < 1e-4$; permutation test; Fig. 4d, inset panel).

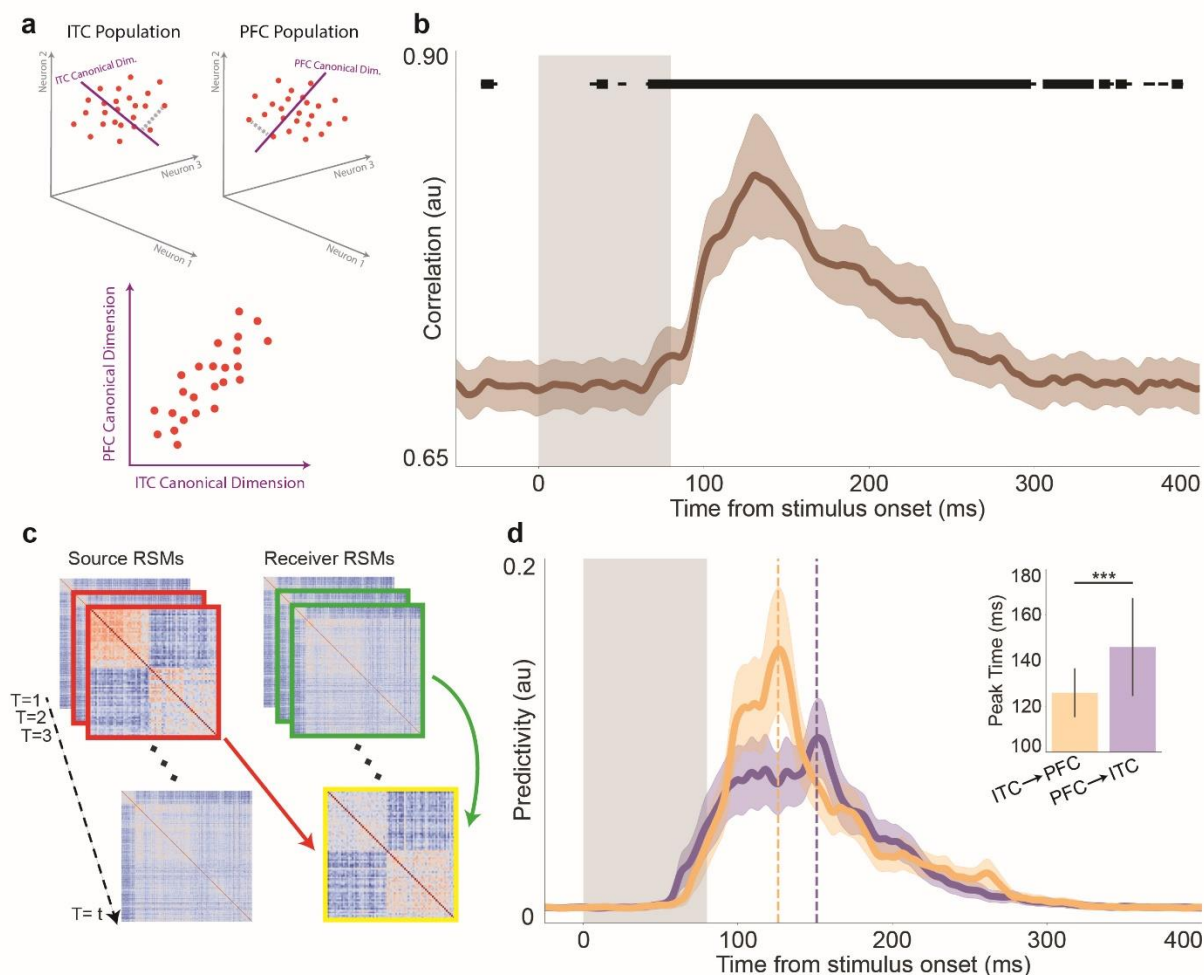


Figure 4. Temporal dynamics of inter-regional population communication in the ITC-PFC circuit.

(a) Schematic of canonical correlation analysis (CCA). CCA identifies dimensions in each neural population space (top) that are maximally correlated across regions (bottom). (b) Time course of population-level functional connectivity between ITC and PFC. Solid lines indicate mean connectivity; shaded areas denote SD. The gray bar (0–80 ms) marks stimulus presentation. Significance was assessed using one-tailed permutation tests; thick and thin horizontal lines indicate $P < 0.001$ and $P < 0.05$ relative to baseline, respectively. (c) Schematic of the Granger causality analysis. The representational similarity matrix (RSM) at time t for a given region (yellow) was predicted from preceding RSMs of the same region (green) or both regions (green and red) using regression models. (d) Time course of Granger-causal influence in the ITC $\rightarrow$ PFC and PFC $\rightarrow$ ITC directions. Solid lines represent mean predictivity; shaded areas denote SD. The gray bar (0–80 ms) indicates stimulus presentation. Dashed lines mark peak latencies of information transfer in each direction. Inset, statistical comparison of peak latencies. Bars indicate medians; error bars denote SD. Significance was assessed using permutation tests (***) $P < 0.001$. CCA: canonical correlation analysis; ITC: inferior temporal cortex; PFC: prefrontal cortex.

This temporal separation is a significant observation, consistent with the PFC feedback being crucial for late-resolved object information^{21,22} and animacy information being extracted at a later phase of visual object processing. This series of evidence strongly suggests that the feed-forward and feedback directions may carry mid- and high-level information, respectively.

Semantic Processing in ITC and PFC Microcircuitries

To explore each region's neural architecture, we used time-time decoding (TTD), which measures the generalization of a representation across time to provide insight into the functional microcircuitry^{44,45}. TTD results for both semantic levels showed an off-diagonal generalization pattern in both regions (Fig. 5a,b; colored pixels denote $p < 0.05$ compared to baseline). This pattern is consistent with a functional structure containing local recurrent interactions^{44–46}.

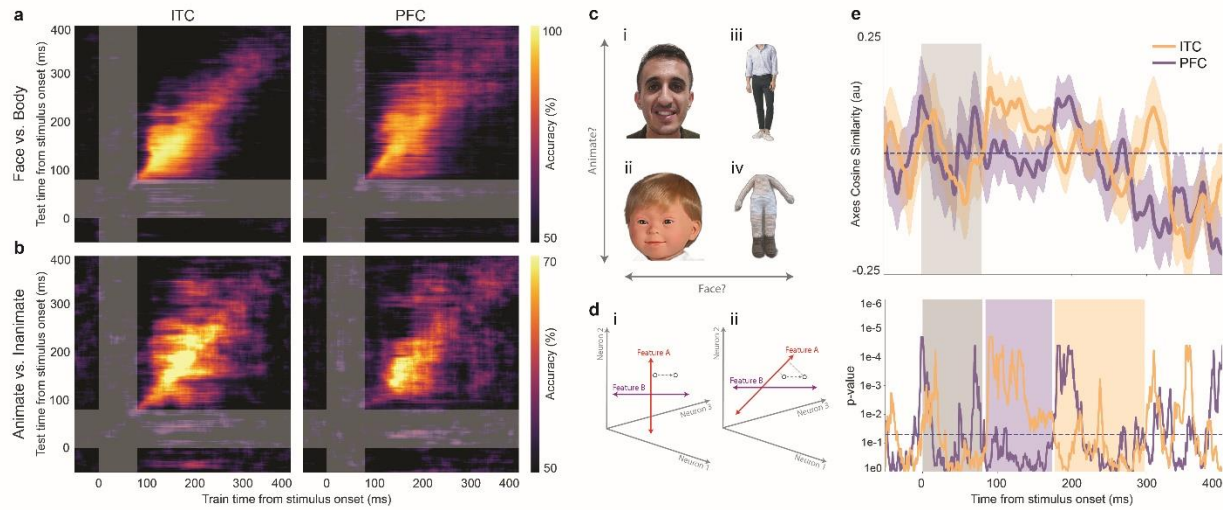


Figure 5. Neural architecture of ITC and PFC microcircuits for semantic processing.

(a,b) Time–time decoding (TTD) matrices for cross-temporal generalization of face–body (a) and animacy (b) information in ITC (left) and PFC (right) populations. The gray bar (0–80 ms) marks stimulus presentation. Significance was assessed using permutation tests relative to baseline; non-significant pixels are shown in black. (c) Schematic of the hypothesized geometric relationship between mid- and high-level information in perceptual space. Animacy (c-i/c-iii versus c-ii/c-iv) is assumed to be independent of face–body status, yielding two orthogonal axes corresponding to the two categorization dimensions. (d) Schematic of possible alignments between feature axes in neural population space. When axes are orthogonal (90°; d-i), variation along one dimension does not affect the other, unlike non-orthogonal configurations (e.g., d-ii). (e, top) Time course of the cosine of the angle between encoding axes corresponding to mid- and high-level abstractions in ITC and PFC. Solid lines indicate mean values; shaded areas denote SD. The gray bar (0–80 ms) indicates stimulus presentation. (e, bottom) Time course of P values for comparison of the between-axis cosine against a null distribution for circular data ($U(-1, 1)$), assessed using permutation tests. The gray bar (0–80 ms) marks stimulus presentation. Colored bars indicate early and late phases of axis orthogonality in PFC and ITC, respectively. ITC: inferior temporal cortex; PFC: prefrontal cortex; TTD: time–time decoding.

Next, we addressed how this network supports perceptual independence for mid- and high-level semantics. Behaviorally, face and animacy judgments are dissociable (Fig. 5c): observers can determine whether a face or a body is animate or inanimate (Fig. 5c, i vs. ii; iii vs. iv), while face perception remains intact for both animate and inanimate objects (Fig. 5c, i vs. iii; i vs. ii). Geometrically, this suggests two orthogonal axes in the perceptual space. We sought to find the neural underpinnings of this independence by defining the encoding axis for each categorization task. To estimate these encoding axes, we used linear discriminant analysis (LDA), a supervised linear dimensionality-reduction method previously applied in similar contexts⁴⁷ (see Methods). If the representations are independent (orthogonal), the cosine will be zero (cosine of 90°) (Fig. 5d).

Fig. 5e (upper panel) demonstrates the temporal dynamics of alignment. Early in processing, the two axes were orthogonally aligned in the PFC neural space (statistical significance is measured by permutation testing against a uniform distribution with a mean at 0; Fig. 5e, lower panel). Subsequently, at the late phase of processing, the same alignment appeared in the ITC population (Fig. 5e). These observations suggest similar encoding mechanisms in ITC and PFC but with distinct timings. The early PFC independence is consistent with an architecture that separates the incoming feed-forward mid-level information from local, ongoing high-level processes into separate subspaces, potentially increasing computational efficiency and providing perceptual independence. The late phase independence in ITC could be analogous to how PFC feedback interacts with concurrent sensory processing during working memory/attention^{39,40}.

Progressive Abstraction in Deep Learning Models

We asked whether the progressive abstraction observed in the primate brain is a general feature of visual systems. We explored semantic representation in state-of-the-art deep learning models of vision (CORnet-S⁴⁸, AlexNet⁴⁹, SqueezeNet⁵⁰, ResNet⁵¹, DenseNet⁵², Inceptionv3⁵³, EfficientNet⁵⁴, VGG-16⁵⁵, and MobileNet⁵⁶- see Methods), which were pretrained for object recognition. Layer-wise RSMs were constructed from the activations and compared to the ground truth (see Methods).

Generally, stronger abstract representations appeared as the input reached the deeper layers of the networks (Fig. 6a,b), for both mid- and high-level abstractions. VGG-16 showed the highest representational similarity for both levels (mean $\pm$ SD of similarity; face-body = 0.62 ± 0.02 , animate-inanimate = 0.21 ± 0.03 ; Fig. 6c,d). These results suggest that, similar to the primate brain, more advanced layers of artificial networks are relatively specialized for more semantic/abstract feature extractions.

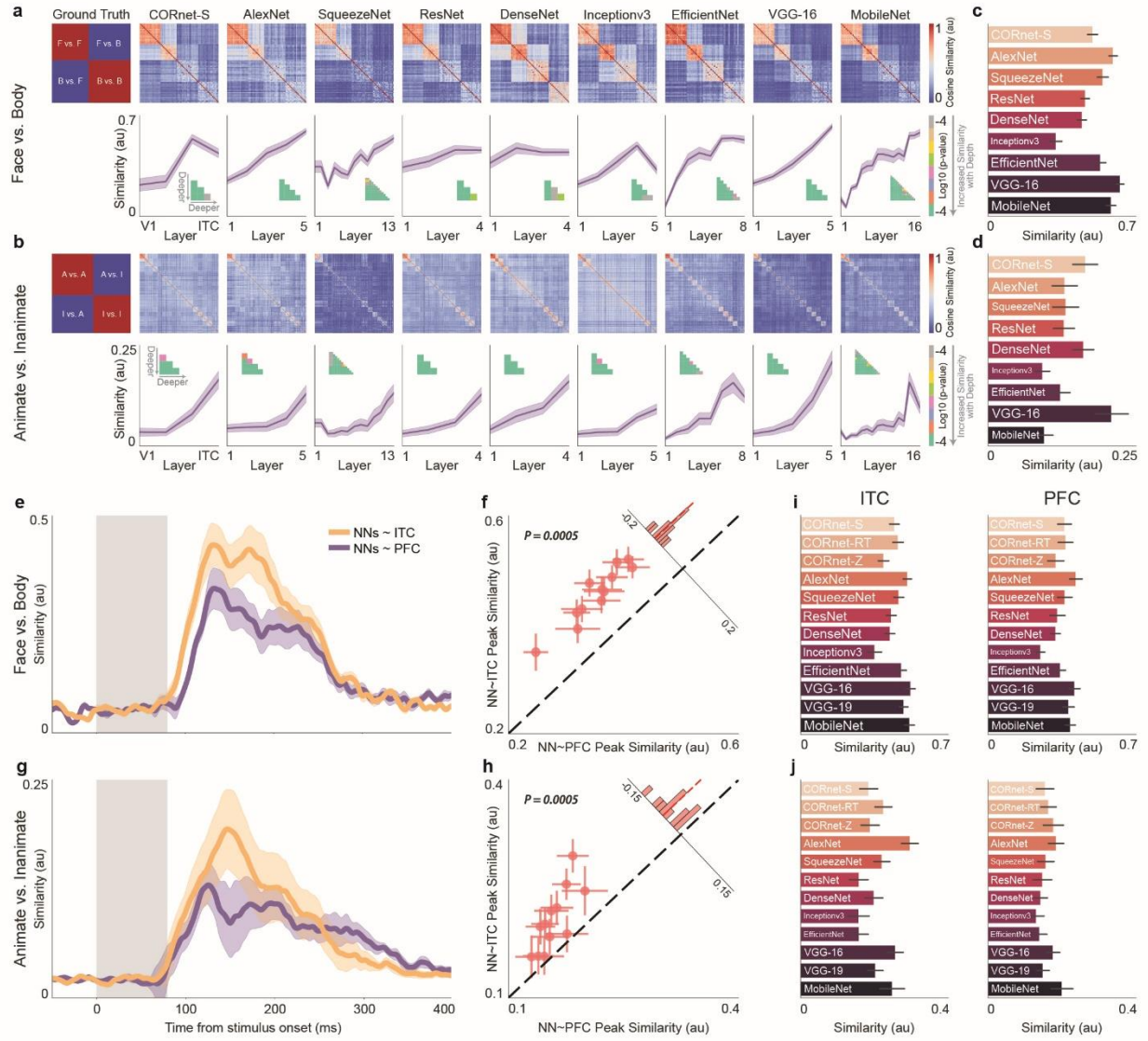


Figure 6. Semantic processing in deep models of vision.

(a,b, top) Representational similarity matrices (RSMs) for ground-truth structure (left) and the final layer of the evaluated networks (right) for mid-level (a; face–body) and high-level (b; animate–inanimate) distinctions. Warmer colors indicate higher cosine similarity. (a,b, bottom) Layer-wise similarity to ground-truth RSMs across all network layers for face–body (a) and animate–inanimate (b) conditions. Solid lines represent mean similarity; shaded areas denote SD. (c,d) Final-layer similarity to ground-truth RSMs across networks for face–body (c) and animate–inanimate (d) distinctions. Bars indicate means; error bars denote SD. (e,g) Time course of similarity between ITC and PFC population representations and the final-layer representations of all networks for face–body (e) and animate–inanimate (g) distinctions. Solid lines indicate mean similarity; shaded areas denote SD. The gray bar (0–80 ms) marks stimulus presentation. (f,h) Scatter plots of peak similarity between each network and ITC versus PFC representations ($n = 12$ networks), corresponding to e and g. Each point represents one network; error bars along axes denote SD. Histograms (top right) show the distribution of similarity differences (NN–ITC versus NN–PFC), with dashed red lines indicating medians. Significance was assessed using two-sided Wilcoxon signed-rank tests against zero difference. (i,j) Peak final-layer similarity to ITC (left) and PFC (right) population representations across networks for face–body

(i) and animate–inanimate (j) distinctions. Bars represent means; error bars denote SD. ITC: inferior temporal cortex; PFC: prefrontal cortex; RSM: representational similarity matrix.

The CORnet family of models is architecturally constrained to reflect the primate ventral visual pathway, with modules corresponding to V1, V2, V4, and ITC⁴⁸. For the analyses in Fig. 6a,b, we used activations from the output layers of these modules in CORnet-S, which shows the highest behavioral and ITC neural predictivity within the family⁴⁸. To assess the similarity between model and neural representations, we used the CORnet-S ITC-equivalent layer as a reference representation for each categorization task (face vs. body; animate vs. inanimate; see Methods). Supplementary Fig. 3a,c shows the time-resolved similarity between CORnet-S representations and population activity in ITC and PFC. CORnet-S ITC-layer representations were significantly more similar to primate ITC than to PFC for both face–body (median $\pm$ SD similarity: ITC = 0.44 ± 0.02 ; PFC = 0.36 ± 0.03 ; $p < 1e-3$, permutation test; Supplementary Fig. 3b) and animate–inanimate (ITC = 0.18 ± 0.02 ; PFC = 0.15 ± 0.02 ; $p < 1e-3$; Supplementary Fig. 3d) distinctions.

Expanding this to other networks (including CORnet-Z, CORnet-RT, VGG-19), Fig. 6e, g shows the temporal evolution of similarity between primate brain representations and those of the deep models. The primate ITC-derived representations were more similar to these networks than PFC representations for both face-body (mean $\pm$ SD of similarity; NN~ITC = 0.46 ± 0.05 , NN~PFC = 0.35 ± 0.04 , $n=12$ networks, $p=0.0005$, one-sample Wilcoxon signed rank test; Fig. 6f) and animate-inanimate (NN~ITC = 0.21 ± 0.04 , NN~PFC = 0.16 ± 0.02 , $n=12$ networks, $p=0.0005$, one-sample Wilcoxon signed rank test; Fig. 6h) cases. VGG-16 showed the highest similarity to ITC (mean $\pm$ SD = 0.52 ± 0.02 ; Fig. 6i, left), while AlexNet was most similar to PFC (mean $\pm$ SD = 0.42 ± 0.03 ; Fig. 6i, right) for face-body. For animacy, AlexNet was most similar to ITC (mean $\pm$ SD = 0.30 ± 0.02 ; Fig. 6j, left), and MobileNet to PFC (mean $\pm$ SD = 0.20 ± 0.03 ; Fig. 6j, right). Overall, these findings are consistent with previous reports^{21,22} and suggest that the visual function of PFC may be partially absent in current deep models. Also, while VGG-16 possesses more visual cortex-like properties, which is similar to a previous study⁵⁷, AlexNet and MobileNet produce more abstract, PFC-like representations.

Discussion

We demonstrate that primate brains process mid-level semantic information in higher visual cortex through feed-forward pathways, while more abstract information is extracted in prefrontal regions, subsequently feeding back to earlier visual areas. Our simultaneous examination of ITC and PFC bridges a critical gap in the literature, linking behavioral evidence of hierarchical processing to the neural circuits that support flexible object recognition. While previous studies have often focused on either ITC or PFC in isolation, our time-resolved analysis provides a more comprehensive view of how sensory features and abstract knowledge interact to yield robust perception. Notably, the process of progressive abstraction is not unique to biological systems, as artificial vision models also exhibit hierarchical processing, albeit with important limitations.

The role of ITC in semantic perception remains a topic of ongoing debate^{12,15,28–30}. Some studies suggest that ITC encodes only physical features^{12–15,20}, with abstraction emerging in downstream regions such as PFC^{8,16}, while others propose that ITC supports categorical representations of evolutionarily relevant classes like faces and bodies^{28–30}. Our findings—showing that mid-level semantic information emerges earlier in ITC and propagates to PFC—support the latter view, highlighting that feed-forward processing alone is insufficient for complete object recognition²². Instead, PFC plays a complementary role by feeding back higher-level information²¹, which aids in resolving more complex categorizations^{9,10,16,31}. The observation that PFC representations of animacy precede those in ITC is consistent with the notion that recurrent ITC–PFC interactions are essential for comprehensive semantic perception. PFC’s ability to integrate information over time and flexibly apply task rules may underlie the efficient "solutions" we identify in object recognition.

Recent studies suggest that core object recognition can largely be explained by feed-forward transformations within the ventral stream, with recurrent processing becoming essential only under more challenging conditions^{22,58}. Other work emphasizes modular, domain-specific organization in high-level visual cortex, proposing that tightly clustered patches in IT support early categorical abstraction^{59,60}. Meanwhile, recent advances have shown that IT population responses lie on smooth, low-dimensional manifolds capable of encoding identity, category, and socially relevant features, suggesting that high-level semantic information may be recoverable directly from the structure of high-level visual representations^{61–63}. Our findings extend these perspectives by demonstrating that, although mid-level semantic structure emerges early in ITC, abstract semantic information is first generated in PFC and subsequently feeds back to ITC. This underscores the idea that ventral-stream representations alone are insufficient for resolving complex semantic distinctions^{22,64,65}. The earlier emergence of high-level semantics in PFC points to a broader, recurrent architecture in which cross-area interactions—rather than ventral-stream mechanisms or IT modularity alone—are key to resolving abstract categories such as animacy^{66–68}.

Whether sequential abstraction is unique to biological systems remains an open question. While deep networks also exhibit progressive abstraction across layers, they lack the recurrent, context-sensitive computations necessary for resolving complex categorizations—precisely where these models often fail^{22,43,69,70}. Incorporating PFC-like mechanisms could enhance the performance of artificial systems. However, fundamental differences in architecture, learning rules, plasticity, and energy consumption suggest that these parallels reflect shared principles rather than mechanistic equivalence^{71,72}.

Our solution time measures reflect passive viewing conditions; however, during active tasks, additional attentional and decision-making demands may alter both the timing and neural pathways underlying semantic resolution^{73,74}. Although we controlled for several low-level visual features, other properties—such as texture, three-dimensional shape, or higher-order regularities—remain potential confounds^{75,76}. Future work using generative modeling or feature disentangling approaches may help isolate the specific semantic representations contributing to solution time. Moreover, while we establish solution time as a candidate neural

metric and relate it to human behavioral literature, we did not collect behavioral measures in macaques. Validating this metric as an index of processing efficiency will require tasks that directly link neural dynamics to reaction times or accuracy. Similarly, although we propose that orthogonal encoding axes in PFC support perceptual independence, this interpretation remains theoretical, and causal manipulations will be necessary to determine whether axis alignment confers a functional perceptual advantage.

Relatedly, our causality analyses should be interpreted with caution. While they reveal predictive relationships between regions, they do not establish direct causal mechanisms or specify the content of feedback signals. One plausible interpretation is that PFC conveys animacy-related information to ITC; however, alternative explanations—including attentional modulation, refinement of visual representations, or the incorporation of prior knowledge—remain viable. Perturbation studies will therefore be essential to adjudicate among these possibilities.

Despite these limitations, our findings are consistent with behavioral evidence supporting sequential perceptual processing, in which mid-level visual features are encoded rapidly, while more abstract properties, such as animacy or identity, emerge later^{7,77,78}. Correspondingly, ITC activity reflects early feature encoding, whereas PFC activity resolves higher-level abstractions following additional processing. This temporal delay likely reflects the time required for information to reach PFC, undergo cognitive transformation, and subsequently influence ITC representations. Importantly, although measures such as onset, peak, and solution time contribute to perceptual outcomes, they may not map directly onto conscious awareness, which depends on additional downstream processes⁷⁹. Thus, while our results align with known human perceptual advantages, they should not be interpreted as direct neural markers of awareness.

In conclusion, recurrent ITC–PFC interactions jointly support semantic perception in primates. This dynamic framework not only accounts for behavioral hierarchies in object recognition but also challenges strictly modular views of visual processing, providing a biologically grounded perspective for developing more adaptive and robust artificial vision systems.

Methods

Animals and Surgery

Two male rhesus macaques (*Macaca mulatta*, monkeys F and V; weights: 9.4 and 8.8 kg; ages: 10 and 9 years) participated in the study. All procedures adhered to the NIH Guide for the Care and Use of Laboratory Animals and were approved by the Royan Institute ethics committee (approval code: IR.ACECR.ROYAN.REC.1399.113). Prior to recordings, each monkey underwent MRI to guide the design and placement of head posts and recording chambers. Head posts were implanted along the midline, and each monkey received a single recording chamber over the left (monkey F) or right (monkey V) hemisphere. In a subsequent surgery, craniotomies were performed to expose regions encompassing both the inferior temporal cortex (ITC) and

ventrolateral prefrontal cortex (vlPFC). Post-surgery CT scans were acquired and co-registered with the MRI to precisely localize the regions of interest.

Stimuli, task, and behavior

Visual stimuli consisted of isolated natural and artificial objects in grayscale, presented on a uniform gray background (Fig. 1a,b). The stimulus set encompassed multiple basic categories to capture the diversity of real-world objects beneath the most abstract level—animacy (Fig. 1b). The animate category included faces and bodies of humans and monkeys, four-limbed animals, reptiles, fishes, birds, and insects, while the inanimate category comprised flowers, fruits, chairs, cars, houses, clocks, and tools. Luminance and spatial frequency profiles were matched across categories using the SHINE toolbox⁸⁰.

Monkeys were trained to maintain central fixation to receive juice rewards, without performing any explicit categorization or identification task. Keeping subjects naïve to category labels minimized the influence of prior experience on the timing of perceptual processing^{7,17} and allowed us to capture the intrinsic representational sequences in the brain. All experiments, including training, were conducted in a single rig, and stimuli were presented using PsychToolbox v3.0.18. Each stimulus appeared centrally (7° visual angle) on a BenQ monitor (1920 × 1080 resolution, 144 Hz refresh rate) at a viewing distance of 50 cm. Eye position was continuously monitored using an infrared eye-tracking system (Zist Kankash Toos, Mashhad, Iran) at 200 Hz. Each stimulus was presented 5–10 times across different recording sessions, with equal repetitions for all stimuli in each session.

Visual feature extraction

Physical features of the objects were extracted using the OpenCV v4.10.0 library in Python. Following preprocessing, the `contourArea`, `arcLength`, and `boundingRect` functions were used to compute the surface area (A_{obj}), perimeter (P_{obj}), and bounding rectangle (i.e., the smallest upright rectangle enclosing all non-background pixels) of each object's contour, respectively.

From these measurements, circularity and elongation were calculated as:

$$Circularity = \frac{4\pi \times A_{obj}}{P_{obj}^2}$$

$$Elongation = 1 - \frac{d_{min}}{d_{max}}$$

where d_{min} and d_{max} are the shorter and longer dimensions of the object's bounding rectangle. Spikiness was computed as:

$$Spikiness = 1 - \frac{A_{obj}}{A_{conv}}$$

where A_{conv} is the area of the object's convex hull, calculated using OpenCV's `convexHull` function.

Luminance and contrast were defined as the mean and standard deviation of the grayscale pixel values, respectively. Additionally, the first two principal components (PC1 and PC2) were computed by performing principal component analysis (PCA) on the grayscale pixel values using the scikit-learn PCA function.

Electrophysiological Recordings

During each session, head-fixed monkeys were seated in a primate chair and viewed visual stimuli presented at the center of the screen. Neural recordings were conducted using custom-designed grids for each monkey's recording chamber, with 1.5 mm and 1 mm spacing between neighboring holes for monkeys F and V, respectively. Tungsten electrodes (FHC, 130 mm; Bowdoin, ME, USA) housed in stainless steel guide tubes were mounted on a Motorized Electrode Manipulator (MEM™; Thomas Recording, Gießen, Hessen, Germany) and carefully advanced through the dura at anterior–posterior (AP) and medial–lateral (ML) coordinates corresponding to the ITC and ventrolateral PFC. After crossing the dura, electrodes were slowly inserted into the brain using the micro-manipulator.

Neural signals were recorded with a Blackrock Neurotech system (Salt Lake City, UT, USA) at a sampling rate of 30 kHz. Across all sessions, 88 recordings were obtained from monkey F and 68 from monkey V. Most sessions included simultaneous recordings from both ITC and PFC, although a few captured only one region. In total, 78 and 59 ITC sites and 57 and 63 PFC sites were recorded from monkeys F and V, respectively. Data from online-detected neural sites with automatic thresholding were used for subsequent analyses.

Neural Data Analysis

Offline analyses were conducted using MATLAB 2022b and Python v3.11.7. Neuronal responses were time-locked to stimulus onset. For each unit, responses were z-scored relative to an 80 ms pre-stimulus baseline and subsequently smoothed using a moving average over consecutive 20 ms windows with 1 ms steps to generate the peri-stimulus time histogram (PSTH). Responses were averaged across all trials for the same stimulus.

For population analyses, each stimulus was represented as a point in an N-dimensional neural space, where N corresponds to the number of recorded neurons in the region. At each timepoint t , the population vector for stimulus i , $S_i(t)$, was defined as:

$$S_i(t) = [r_1(t), r_2(t), \dots, r_N(t)]$$

where $r_j(t)$ is the response of neuron j to stimulus i at time t .

Response onset was defined as the first timepoint at which the PSTH exceeded a threshold, calculated as the mean baseline activity plus three times the standard deviation of the baseline. The baseline was defined as the period from –50 to 50 ms relative to stimulus onset. Repetitions (for population analyses)

and units (for encoding models) with onset or peak times outside the window of 50–300 ms post-stimulus were considered unreliable and excluded from further analyses. Solution time was calculated as the difference between onset and peak times.

Classification and time-time decoding

Population-level neural data from ITC and PFC were analyzed using linear discriminant analysis (LDA; MATLAB functions `fitcdiscr` and `predict`) to decode representations at different semantic levels (Fig. 1g–j, Fig. 5a,b). For standard classification, an LDA classifier was trained at each timepoint to discriminate either animates versus inanimates (high-level abstraction) or faces versus bodies (mid-level abstraction). Data were split into 70% for training and 30% for testing. Confusion matrices were computed for each iteration, and the mean within-class accuracy was used as the performance metric. This procedure was repeated 200 times to ensure reliability. Onset, peak, and solution times were subsequently computed as described above.

For time–time decoding (TTD; Fig. 5a,b), classifiers were trained at each timepoint on mid- and high-level distinctions and then tested across all timepoints to assess the temporal generalization of neural representations^{44,45}. Each TTD analysis was repeated over 100 iterations, with a 70%/30% train/test split for each run.

Representational similarity analysis

Representational similarity analysis (RSA)²⁷ was performed to quantify the similarity of neural population responses across stimuli. For each semantic condition, a ground-truth representational similarity matrix (RSM) was first constructed, where entries were 0 for dissimilar stimuli and 1 for identical or perfectly similar stimuli.

At each timepoint and for each region, the cosine similarity between neural response vectors for all pairs of stimuli, $Similarity_{i,j}$, was computed:

$$Similarity_{i,j} = \cos(\theta_{i,j}) = \frac{Resp_{stim_i} \cdot Resp_{stim_j}}{\|Resp_{stim_i}\| \|Resp_{stim_j}\|}$$

where $Resp_{stim_i}$ and $Resp_{stim_j}$ are the population vectors of neural responses to stimuli i and j , and $\theta_{i,j}$ is the angle between these vectors in high-dimensional neural space. Only responses to face and body stimuli were used to construct the face–body RSMs. This procedure generated instantaneous $N \times N$ RSMs for each region and condition, where N is the total number of stimuli and $Similarity_{i,j}$ represents the cosine similarity between stimuli i and j for face-body and animate-inanimate conditions.

To quantify correspondence with the ground truth, each data-derived RSM was correlated with its respective ground-truth RSM using Kendall’s tau. This process was repeated 200 times, and for each iteration, a subset of stimuli (20/50 per class; total 40/100 for face–body, or 40/100 for

animate–inanimate) was randomly selected. Onset, peak, and solution times were computed as described above.

Generalized linear models

Encoding models, specifically GLMs, were formed to predict the neural response from a set of semantic and physical features of each object. Specifically, the semantic features were the following variables in binary format: animate, face, body, human, monkey; each regressor for each object was either True or False. Physical features were circularity, elongation, spikiness, contrast, luminance, object area as well as the first and second principal components of the image in a grayscale, as floating-point numbers. One model was formed for every unit at every timepoint (using Statsmodels *GLM* function). The full model was as the following:

$$y_t = \beta_1 \times X_{animacy} + \beta_2 \times X_{face} + \beta_3 \times X_{body} + \beta_4 \times X_{human} + \beta_5 \times X_{monkey} + \beta_6 \times X_{circularity} + \beta_7 \times X_{elongation} + \beta_8 \times X_{spikiness} + \beta_9 \times X_{contrast} \\ + \beta_{10} \times X_{PC1} + \beta_{11} \times X_{PC2} + \beta_{12} \times X_{luminance} + \beta_{13} \times X_{object_area}$$

where y_t is the baseline z-scored, stimulus-averaged, and smoothed (moving average with 20-ms window and 1-ms step) neural response at timepoint t . Next, similar to the notion of a recent study³³, which states:

$$neural\ firing\ rate = firing\ rates\ explained\ by\ regressors + residuals$$

and is because:

$$total\ sum\ of\ squares\ (TSS) \\ = explained\ sum\ of\ squares\ (ESS) + residual\ sum\ of\ squares\ (RSS)$$

Or

$$\sum_{i=1}^n (y_i - \underline{y})^2 = \sum_{i=1}^n (\hat{y}_i - \underline{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

in which y_i and $\hat{y}_i$ are the true and model-predicted neural response to stimuli, respectively, and $\underline{y}$ is the average neural response to all stimuli, to extract object residuals when animacy is absent in the model (and therefore present and enhanced in the residuals), we formed models without the $\beta_1 \times X_{animacy}$ term and computed the residuals as:

$$residual_i(t) = y_i(t) - \hat{y}_i(t)$$

where, at timepoint t , $residual_i(t)$ is the model's residual for $stimulus_i$, $y_i(t)$ is the neural response of a given neuron to the same stimulus, and $\hat{y}_i(t)$ is the model prediction of the neural response of that particular neuron and stimulus. This procedure was performed for all neurons of the population. Next, the computed residuals, that are free of all other regressors and have enhanced animacy signals, were normalized to the baseline and entered the RSA for 500 times, similar to the process described above. The same method was applied to enhance the face and

body signals; this time, the $\beta_2 \times X_{face}$ and $\beta_3 \times X_{body}$ terms were excluded from the initial predictive model. Subsequently, the onset, peak, and solution times were computed as described above.

Canonical correlation analysis

To measure the functional connectivity between two neuronal populations, we used CCA⁸¹, similar to a recent study³⁵. Briefly, CCA defines pairs of dimensions in the neural spaces of two areas (will be denoted as a, b), one in each area, that meets the following criterion: once the neural activities of the two regions are projected onto the corresponding dimensions, they give maximum possible correlation; specifically, at every timepoint we have:

$$(a, b) = \arg_{a,b} \text{corr}(Xa, Yb)$$

where X and Y are the $n \times p_x$ and $n \times p_y$ matrices of residual activities in the two areas, respectively, in which n is the number of data points and p_x and p_y are the number of recorded neurons in the two regions, respectively. The vectors a and b have dimensions $p_x \times 1$ and $p_y \times 1$, respectively.

Granger causality

To quantify information transfer rate over time, we used the RSMs created during the RSA (see above), in a way similar to a previous study^{41–43}. First, the RSMs were created for each region separately over time; this would result in $N \times N \times T$ matrices for each region, where N is the number of stimuli and T is number of timepoints. Next, we formed two regression models to predict the RSM at timepoint t (RSM_t) of the theoretical receiver area from the RSMs of timepoints $[t - 60, t - 20]$ in the following manner: (1) the first regression model only had the past RSMs of the receiver area to predict the RSM_t , while (2) the second model had to predict the RSM_t from the past RSMs of both areas. Hypothetically, if the theoretical sender area transfers information to the theoretical receiver area the residual of the second model should be lower than that of the first model. To compare the residuals, we used the following formulation:

$$\text{Predictivity} = \log \log \frac{SSR_{\text{model}(2)}}{SSR_{\text{model}(1)}}$$

where $SSR_{\text{model}(1)}$ and $SSR_{\text{model}(2)}$ are the sum squared residuals for model (1) and model (2), respectively. PFC and ITC were the receiver areas for the feedforward and feedback directions, respectively. This procedure was repeated 200 times, and in every run, 100 stimuli were randomly selected to measure the information transfer. Subsequently, the peak information transfer was calculated for each direction.

Encoding axes alignment

To identify the animacy- and face–body–specific encoding axes in ITC and PFC, we employed a linear discriminant analysis (LDA) classifier using the scikit-learn *LinearDiscriminantAnalysis* function⁴⁷. LDA is a supervised dimensionality reduction method that maximizes the following objective function:

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

where S_B is the between-class scatter matrix (quantifying the squared distance between class means), S_W is the within-class scatter matrix (quantifying the variance within each class), and w is the weight vector. Each element of w corresponds to the contribution of an individual neuron to the discrimination task. Maximizing $J(w)$ ensures that the projected class means are maximally separated relative to within-class variance, yielding a high-dimensional neural axis representing task-specific encoding.

At each timepoint and in each region, the LDA classifier was trained to discriminate animate from inanimate objects (or faces from bodies). The resulting weight vector was extracted and considered the animacy or face–body encoding axis. To assess the relationship between axes, the cosine of the angle between the animacy and face–body axes was computed. Model fitting used the eigenvalue decomposition solver, with the shrinkage parameter automatically determined via the Ledoit–Wolf lemma⁸². This procedure was repeated 200 times, using 70% of samples to train the model in each iteration.

Deep learning models of vision

Neural network evaluations were conducted using PyTorch v2.3.0 and Torchvision v0.18.0. Layers of interest for each network are listed in Table 1. For each network, activations of the selected layers were first extracted in a purely feedforward manner. Layer-wise representational similarity matrices (RSMs) were then computed and compared to the corresponding ground truth matrices using the same procedure applied to neural data (Fig. 6a–d).

To quantify brain–network similarity (Fig. 6e–j, Supplementary Fig. 3), RSMs from the final layer of interest in each network were compared against neural RSMs instead of the theoretical ground truth matrices used in Fig. 2c,g. All procedures were repeated 200 times. In each iteration, a subset of stimuli—20/50 per class, totaling 40/100 objects—was randomly selected for the face–body or animate–inanimate conditions. Peak similarity values from each repetition were used for subsequent statistical analyses.

Table 1. Layers of interest for deep models of vision

Network	Layers
CORnet-S CORnet-RT CORnet-Z	"module.V1.output", "module.V2.output", "module.V4.output", "module.IT.output"
AlexNet	"features.1", "features.4", "features.7", "features.9", "features.12"
SqueezeNet	"features.0", "features.1", "features.2", "features.3.cat", "features.4.cat", "features.5", "features.6.cat", "features.7.cat", "features.8", "features.9.cat", "features.10.cat", "features.11.cat", "features.12.cat"
ResNet	"layer1.2.relu_2", "layer2.3.relu_2", "layer3.5.relu_2", "layer4.2.relu_2"
DenseNet	"features.transition1.pool", "features.transition2.pool", "features.transition3.pool", "flatten"
Inceptionv3	"maxpool1", "maxpool2", "Mixed_5d.cat", "Mixed_6e.cat", "avgpool"
EfficientNet	"features.1.1.add", "features.2.2.add", "features.3.2.add", "features.4.3.add", "features.5.3.add", "features.6.4.add", "features.7.1.add", "avgpool"
VGG-16	"features.4", "features.9", "features.16", "features.23", "features.30"
VGG-19	"features.4", "features.9", "features.18", "features.27", "features.36"
MobileNet	"features.0", "features.1.add", "features.2.block.2", "features.3.add", "features.4.block.3", "features.5.add", "features.6.add", "features.7.block.2", "features.8.add", "features.9.add", "features.10.add", "features.11.block.3", "features.12.add", "features.13.block.3", "features.14.add", "features.15.add", "features.16"

Statistical analyses

Statistical and machine learning analyses were performed in Python v3.11.7, using scikit-learn v1.2.2, Statsmodels v0.14.0, and SciPy v1.13.0 libraries, and MATLAB 2022b. Details of the statistical tests used for each comparison are described wherever appropriate throughout the text. All permutations were repeated 100001 times, except for Fig. 5A,B (1001 times). All tests were two-tailed, unless mentioned otherwise, and p-values less than 0.05 were considered as statistically significant.

References

1. DiCarlo, J. J. & Cox, D. D. Untangling invariant object recognition. *Trends Cogn. Sci.* **11**, 333–341 (2007).
2. DiCarlo, J. J., Zoccolan, D. & Rust, N. C. How does the brain solve visual object recognition? *Neuron* **73**, 415–434 (2012).
3. Serre, T., Oliva, A. & Poggio, T. A feedforward architecture accounts for rapid categorization. *Proceedings of the national academy of sciences* **104**, 6424–6429 (2007).

4. Mack, M. L. & Palmeri, T. J. The timing of visual object categorization. *Front. Psychol.* **2**, 165 (2011).
5. Mack, M. L. & Palmeri, T. J. The dynamics of categorization: Unraveling rapid categorization. *J. Exp. Psychol. Gen.* **144**, 551 (2015).
6. Riesenhuber, M. & Poggio, T. Models of object recognition. *Nat. Neurosci.* **3**, 1199–1204 (2000).
7. Dehaqani, M.-R. A. *et al.* Temporal dynamics of visual category representation in the macaque inferior temporal cortex. *J. Neurophysiol.* **116**, 587–601 (2016).
8. Meyers, E. M., Freedman, D. J., Kreiman, G., Miller, E. K. & Poggio, T. Dynamic population coding of category information in inferior temporal and prefrontal cortex. *J. Neurophysiol.* **100**, 1407–1419 (2008).
9. Rainer, G., Rao, S. C. & Miller, E. K. Prospective coding for objects in primate prefrontal cortex. *Journal of Neuroscience* **19**, 5493–5505 (1999).
10. Wallis, J. D., Anderson, K. C. & Miller, E. K. Single neurons in prefrontal cortex encode abstract rules. *Nature* **411**, 953–956 (2001).
11. Lehky, S. R. & Tanaka, K. Neural representation for object recognition in inferotemporal cortex. *Curr. Opin. Neurobiol.* **37**, 23–35 (2016).
12. Vinken, K., Prince, J. S., Konkle, T. & Livingstone, M. S. The neural code for “face cells” is not face-specific. *Sci. Adv.* **9**, eadg1736 (2023).
13. Bardon, A., Xiao, W., Ponce, C. R., Livingstone, M. S. & Kreiman, G. Face neurons encode nonsemantic features. *Proceedings of the national academy of sciences* **119**, e2118705119 (2022).
14. Arcaro, M. J., Ponce, C. & Livingstone, M. The neurons that mistook a hat for a face. *Elife* **9**, e53798 (2020).
15. Baldassi, C. *et al.* Shape similarity, better than semantic membership, accounts for the structure of visual object representations in a population of monkey inferotemporal neurons. *PLoS Comput. Biol.* **9**, e1003167 (2013).
16. Miller, E. K. & Cohen, J. D. An integrative theory of prefrontal cortex function. *Annu. Rev. Neurosci.* **24**, 167–202 (2001).
17. Tanaka, J. W. & Taylor, M. Object categories and expertise: Is the basic level in the eye of the beholder? *Cogn. Psychol.* **23**, 457–482 (1991).
18. Johnson, K. E. & Mervis, C. B. Effects of varying levels of expertise on the basic level of categorization. *J. Exp. Psychol. Gen.* **126**, 248 (1997).
19. Freedman, D. J., Riesenhuber, M., Poggio, T. & Miller, E. K. Categorical representation of visual stimuli in the primate prefrontal cortex. *Science* (1979). **291**, 312–316 (2001).
20. Sharma, S., Vinken, K., Jagadeesh, A. V & Livingstone, M. S. Face cells encode object parts more than facial configuration of illusory faces. *Nat. Commun.* **15**, 9879 (2024).

21. Kar, K. & DiCarlo, J. J. Fast recurrent processing via ventrolateral prefrontal cortex is needed by the primate ventral stream for robust core visual object recognition. *Neuron* **109**, 164–176 (2021).
22. Kar, K., Kubilius, J., Schmidt, K., Issa, E. B. & DiCarlo, J. J. Evidence that recurrent circuits are critical to the ventral stream's execution of core object recognition behavior. *Nat. Neurosci.* **22**, 974–983 (2019).
23. Toosi, R. *et al.* The spatial frequency representation predicts category coding in the inferior temporal cortex. *Elife* **13**, RP93589 (2025).
24. Emadi, N. & Esteky, H. Neural representation of ambiguous visual objects in the inferior temporal cortex. *PLoS One* **8**, e76856 (2013).
25. Nosofsky, R. M. & Palmeri, T. J. Comparing exemplar-retrieval and decision-bound models of speeded perceptual classification. *Percept. Psychophys.* **59**, 1027–1048 (1997).
26. Nosofsky, R. M. & Palmeri, T. J. An exemplar-based random walk model of speeded classification. *Psychol. Rev.* **104**, 266 (1997).
27. Kriegeskorte, N., Mur, M. & Bandettini, P. A. Representational similarity analysis-connecting the branches of systems neuroscience. *Front. Syst. Neurosci.* **2**, 249 (2008).
28. Shi, Y. *et al.* Rapid, concerted switching of the neural code in inferotemporal cortex. *bioRxiv* 2012–2023 (2023).
29. Landi, S. M., Viswanathan, P., Serene, S. & Freiwald, W. A. A fast link between face perception and memory in the temporal pole. *Science* (1979). **373**, 581–585 (2021).
30. Kriegeskorte, N. *et al.* Matching categorical object representations in inferior temporal cortex of man and monkey. *Neuron* **60**, 1126–1141 (2008).
31. Miller, E. K. The prefrontal cortex and cognitive control. *Nat. Rev. Neurosci.* **1**, 59–65 (2000).
32. Rose, O. & Ponce, C. R. A concentration of visual cortex-like neurons in prefrontal cortex. *Nat. Commun.* **15**, 7002 (2024).
33. Yu, G., Katz, L. N., Quaia, C., Messinger, A. & Krauzlis, R. J. Short-latency preference for faces in primate superior colliculus depends on visual cortex. *Neuron* **112**, 2814–2822 (2024).
34. Noroozi, J., Rezayat, E. & Dehaqani, M.-R. A. Frontotemporal network contribution to occluded face processing. *Proceedings of the National Academy of Sciences* **121**, e2407457121 (2024).
35. Semedo, J. D. *et al.* Feedforward and feedback interactions between visual cortical areas use different population activity patterns. *Nat. Commun.* **13**, 1099 (2022).
36. Seger, C. A. & Miller, E. K. Category learning in the brain. *Annu. Rev. Neurosci.* **33**, 203–219 (2010).
37. Webster, M. J., Bachevalier, J. & Ungerleider, L. G. Connections of inferior temporal areas TEO and TE with parietal and frontal cortex in macaque monkeys. *Cerebral cortex* **4**, 470–483 (1994).
38. Ungerleider, L. G., Gaffan, D. & Pelak, V. S. Projections from inferior temporal cortex to prefrontal cortex via the uncinate fascicle in rhesus monkeys. *Exp. Brain Res.* **76**, 473–484 (1989).

39. Buschman, T. J. Balancing flexibility and interference in working memory. *Annu. Rev. Vis. Sci.* **7**, 367–388 (2021).
40. Stokes, M. G. *et al.* Dynamic coding for cognitive control in prefrontal cortex. *Neuron* **78**, 364–375 (2013).
41. Granger, C. W. J. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* 424–438 (1969).
42. Bernasconi, C. & KoÈnig, P. On the directionality of cortical interactions studied by structural analysis of electrophysiological recordings. *Biol. Cybern.* **81**, 199–210 (1999).
43. Kietzmann, T. C. *et al.* Recurrence is required to capture the representational dynamics of the human visual system. *Proceedings of the National Academy of Sciences* **116**, 21854–21863 (2019).
44. King, J.-R. & Dehaene, S. Characterizing the dynamics of mental representations: the temporal generalization method. *Trends Cogn. Sci.* **18**, 203–210 (2014).
45. Rajaei, K., Mohsenzadeh, Y., Ebrahimpour, R. & Khaligh-Razavi, S.-M. Beyond core object recognition: Recurrent processes account for object recognition under occlusion. *PLoS Comput. Biol.* **15**, e1007001 (2019).
46. King, J.-R., Pescetelli, N. & Dehaene, S. Brain mechanisms underlying the brief maintenance of seen and unseen sensory information. *Neuron* **92**, 1122–1134 (2016).
47. Dehaqani, A. A. *et al.* A mechanosensory feedback that uncouples external and self-generated sensory responses in the olfactory cortex. *Cell Rep.* **43**, (2024).
48. Kubilius, J. *et al.* Brain-like object recognition with high-performing shallow recurrent ANNs. *Adv. Neural Inf. Process. Syst.* **32**, (2019).
49. Krizhevsky, A., Sutskever, I. & Hinton, G. E. ImageNet classification with deep convolutional neural networks. *Commun. ACM* **60**, 84–90 (2017).
50. Iandola, F. N. *et al.* SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and < 0.5 MB model size. *arXiv preprint arXiv:1602.07360* (2016).
51. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. in *Proceedings of the IEEE conference on computer vision and pattern recognition* 770–778 (2016).
52. Huang, G., Liu, Z., Van Der Maaten, L. & Weinberger, K. Q. Densely connected convolutional networks. in *Proceedings of the IEEE conference on computer vision and pattern recognition* 4700–4708 (2017).
53. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J. & Wojna, Z. Rethinking the inception architecture for computer vision. in *Proceedings of the IEEE conference on computer vision and pattern recognition* 2818–2826 (2016).
54. Tan, M. & Le, Q. Efficientnet: Rethinking model scaling for convolutional neural networks. in *International conference on machine learning* 6105–6114 (PMLR, 2019).

55. Simonyan, K. & Zisserman, A. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556* (2014).
56. Howard, A. G. *et al.* Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861* (2017).
57. Nonaka, S., Majima, K., Aoki, S. C. & Kamitani, Y. Brain hierarchy score: Which deep neural networks are hierarchically brain-like? *iScience* **24**, (2021).
58. Kar, K. & DiCarlo, J. J. The quest for an integrated set of neural mechanisms underlying object recognition in primates. *Annu. Rev. Vis. Sci.* **10**, (2024).
59. Arcaro, M. & Livingstone, M. A whole-brain topographic ontology. *Annu. Rev. Neurosci.* **47**, (2024).
60. Livingstone, M. S., Arcaro, M. J. & Schade, P. F. Cortex is cortex: Ubiquitous principles drive face-domain development. *Trends Cogn. Sci.* **23**, 3–4 (2019).
61. Bao, P., She, L., McGill, M. & Tsao, D. Y. A map of object space in primate inferotemporal cortex. *Nature* **583**, 103–108 (2020).
62. Chang, L., Egger, B., Vetter, T. & Tsao, D. Y. Explaining face representation in the primate brain using different computational models. *Current Biology* **31**, 2785–2795 (2021).
63. Zhuang, C. *et al.* Unsupervised neural network models of the ventral visual stream. *Proceedings of the National Academy of Sciences* **118**, e2014196118 (2021).
64. Bonnen, T., Yamins, D. L. K. & Wagner, A. D. When the ventral visual stream is not enough: A deep learning account of medial temporal lobe involvement in perception. *Neuron* **109**, 2755–2766 (2021).
65. Deniz, F., Nunez-Elizalde, A. O., Huth, A. G. & Gallant, J. L. The representation of semantic information across human cerebral cortex during listening versus reading is invariant to stimulus modality. *Journal of Neuroscience* **39**, 7722–7736 (2019).
66. Zhou, T. *et al.* Mapping information flow between the inferotemporal and prefrontal cortices via neural oscillations in memory retrieval and maintenance. *Cell Rep.* **42**, (2023).
67. Von Seth, J., Nicholls, V. I., Tyler, L. K. & Clarke, A. Recurrent connectivity supports higher-level visual and semantic object representations in the brain. *Commun. Biol.* **6**, 1207 (2023).
68. Von Seth, J., Nicholls, V. I., Tyler, L. K. & Clarke, A. Recurrent connectivity supports higher-level visual and semantic object representations in the brain. *Commun. Biol.* **6**, 1207 (2023).
69. Yamins, D. L. K. & DiCarlo, J. J. Using goal-driven deep learning models to understand sensory cortex. *Nat. Neurosci.* **19**, 356–365 (2016).
70. Rajalingham, R. *et al.* Large-scale, high-resolution comparison of the core visual object recognition behavior of humans, monkeys, and state-of-the-art deep artificial neural networks. *Journal of Neuroscience* **38**, 7255–7269 (2018).
71. Lillicrap, T. P., Santoro, A., Marris, L., Akerman, C. J. & Hinton, G. Backpropagation and the brain. *Nat. Rev. Neurosci.* **21**, 335–346 (2020).

72. Marblestone, A. H., Wayne, G. & Kording, K. P. Toward an integration of deep learning and neuroscience. *Front. Comput. Neurosci.* **10**, 215943 (2016).
73. Toneva, M., Stretcu, O., Póczos, B., Wehbe, L. & Mitchell, T. M. Modeling task effects on meaning representation in the brain via zero-shot meg prediction. *Adv. Neural Inf. Process. Syst.* **33**, 5284–5295 (2020).
74. Moisa, M. *et al.* Brain activity during divided and selective attention to auditory and visual sentence comprehension tasks. *Front. Hum. Neurosci.* **9**, 86 (2015).
75. Ullman, S., Assif, L., Fetaya, E. & Harari, D. Atoms of recognition in human and computer vision. *Proceedings of the National Academy of Sciences* **113**, 2744–2749 (2016).
76. Kourtzi, Z. & Kanwisher, N. Representation of perceived object shape by the human lateral occipital complex. *Science* (1979). **293**, 1506–1509 (2001).
77. Mack, M. L., Wong, A. C.-N., Gauthier, I., Tanaka, J. W. & Palmeri, T. J. Time course of visual object categorization: Fastest does not necessarily mean first. *Vision Res.* **49**, 1961–1968 (2009).
78. Wang, R., Janini, D. & Konkle, T. Mid-level feature differences support early animacy and object size distinctions: Evidence from electroencephalography decoding. *J. Cogn. Neurosci.* **34**, 1670–1680 (2022).
79. Blumenfeld, H. Brain mechanisms of conscious awareness: detect, pulse, switch, and wave. *The Neuroscientist* **29**, 9–18 (2023).
80. Willenbockel, V. *et al.* Controlling low-level image properties: the SHINE toolbox. *Behav. Res. Methods* **42**, 671–684 (2010).
81. Hotelling, H. Relations between two sets of variates. in *Breakthroughs in statistics: methodology and distribution* 162–190 (Springer, 1992).
82. Ledoit, O. & Wolf, M. A well-conditioned estimator for large-dimensional covariance matrices. *J. Multivar. Anal.* **88**, 365–411 (2004).

Author Contributions

MRAD, FS and MZ conceptualized and designed the study. FS created the stimulus set. MR, MZ, MQ, and MM collected the data. MRAD and MM designed the analysis plan, analyzed the data. MM performed neural network evaluations, and generated visualizations. MM and FS wrote the manuscript. FS, MJ, and MRAD supervised the study.

Data Availability

Data will be made available upon reasonable request to the corresponding author.

Code Availability

MATLAB scripts and functions as well as Python notebooks will be made publicly available at https://github.com/mooziri/Paper_VisualSemanticProcessing following publication of the study.

Competing Interests

No conflict. The authors declare no competing financial interests.

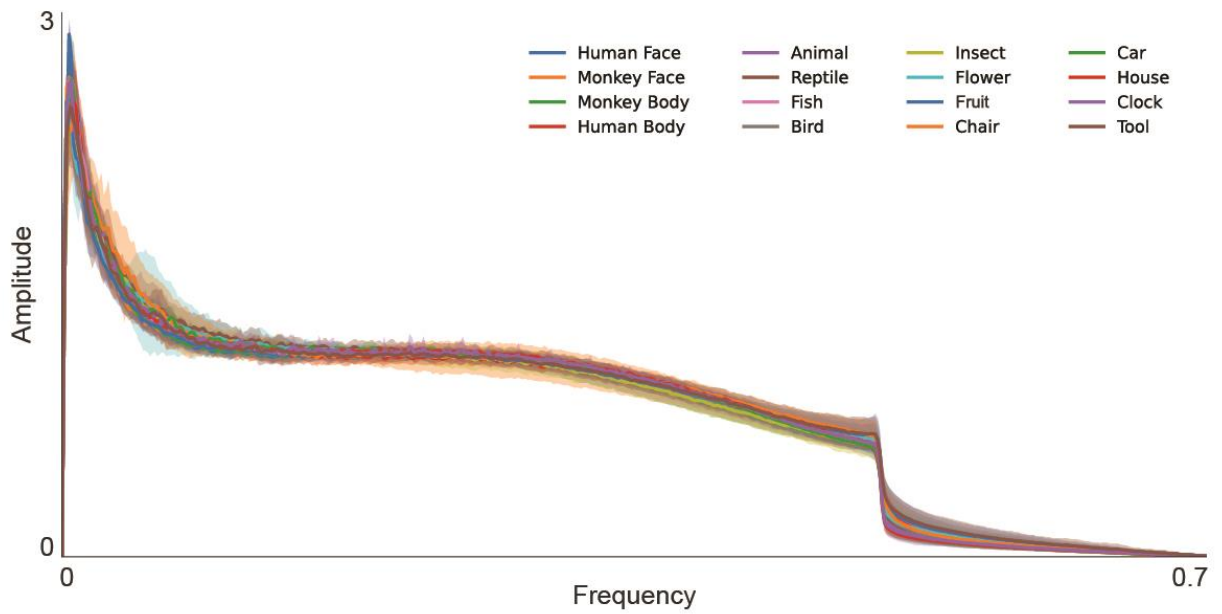
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During the preparation of this manuscript, the authors used AI tools to assist with language editing, grammar checking, and rephrasing selected sections for clarity. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the final version of the paper.

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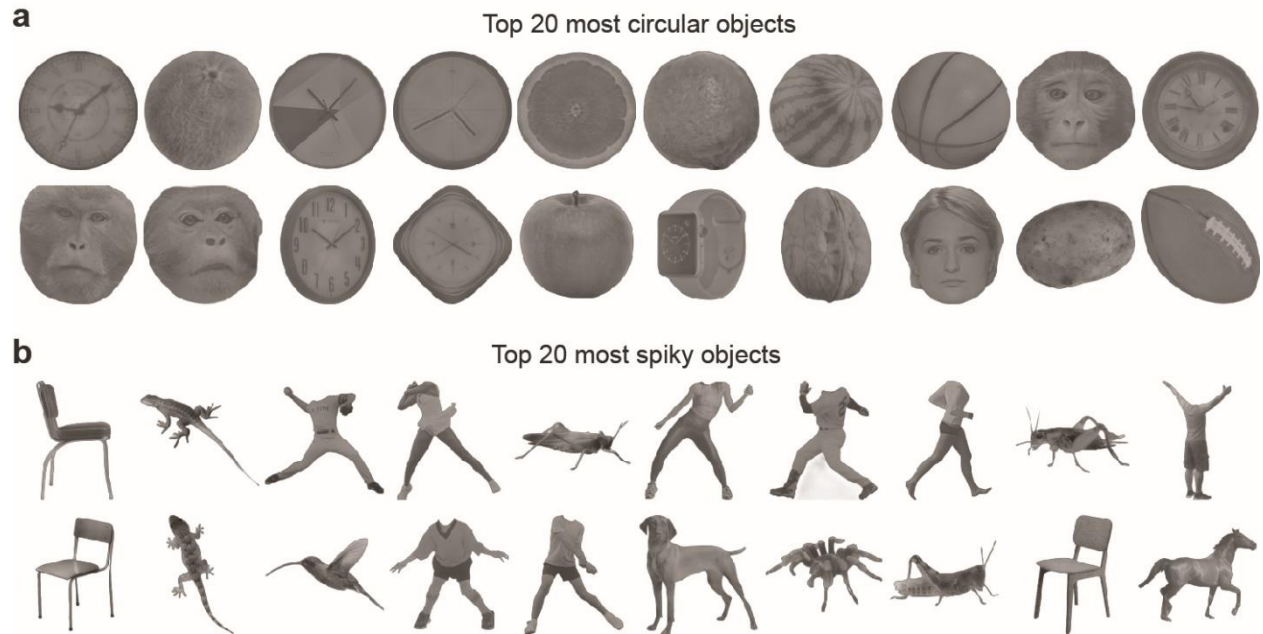
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Supplementary Information



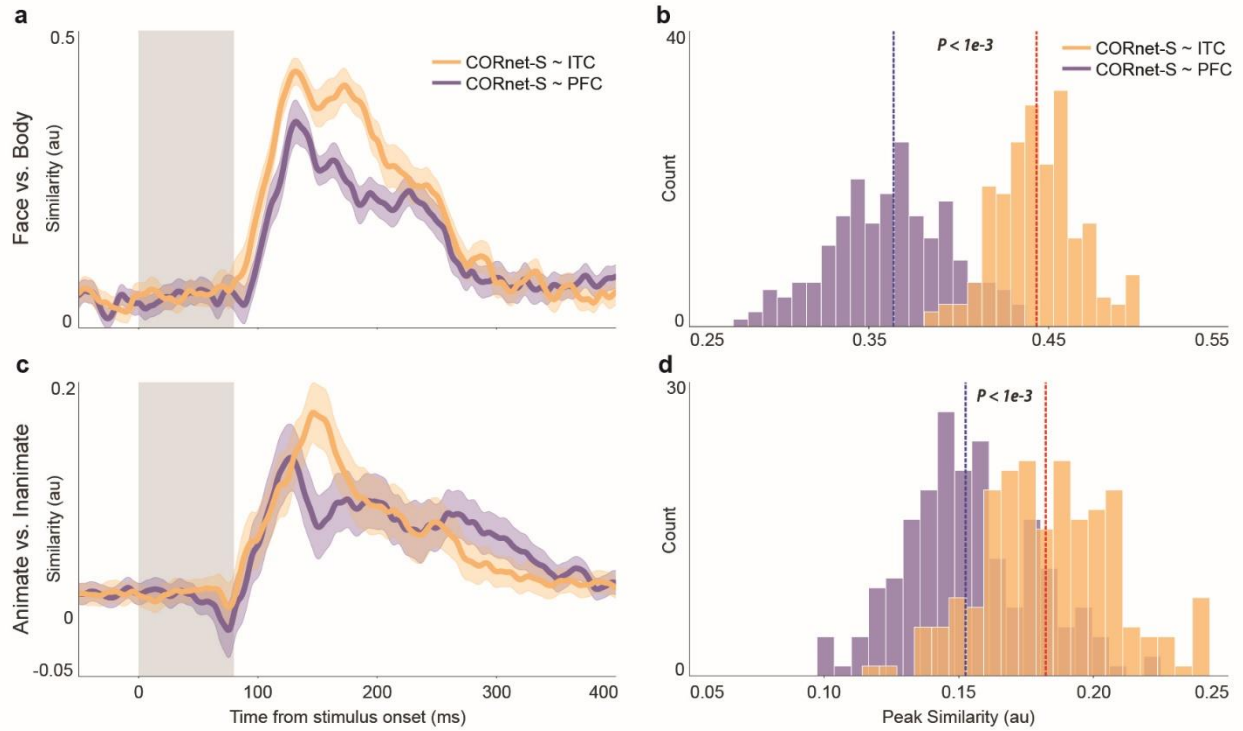
Supplementary Figure 1. Matching spatial frequency across categories.

Spatial frequency spectra of visual stimuli for each category. Solid lines indicate mean amplitude; shaded areas denote SD. Statistical significance was assessed using Mann–Whitney tests between category pairs at each spatial frequency.



Supplementary Figure 2. Visual feature differences across high-level semantic categories.

(a,b) Examples of stimuli with extreme shape properties. Shown are the 20 most circular **(a)** and 20 most spiky **(b)** objects from the full stimulus set.



Supplementary Figure 3. Semantic similarity between CORnet-S and primate ITC and PFC.

(a,c) Time course of similarity to the CORnet-S ITC layer for population activity in ITC and PFC for face–body (A) and animate–inanimate (c) distinctions. Solid lines indicate mean similarity; shaded areas denote SD. The gray bar (0–80 ms) marks stimulus presentation. (b,d) Distributions of peak similarity values corresponding to a and c. Dashed lines indicate medians. Significance was assessed using permutation tests. ITC: inferior temporal cortex; PFC: prefrontal cortex.